

Question number	Answer	Additional guidance	Mark
1(a)	An answer that includes two of the following points: - sterilise equipment (1) - work beside a bunsen burner (1) - cover / minimise the length of time that cultures are open (1)	ACCEPT named methods of sterilisation and named equipment washing hands / gloves / wiping down surfaces with disinfectant ACCEPT selective media NB aseptic technique if no other marks awarded	(2)

Question number	Answer	Mark
1(b)	The only correct answer is C <i>A is incorrect because 10 µg is 0.01 mg, and 0.5 mg ÷ 0.01 = 50</i> <i>B is incorrect because 10 µg is 0.01 mg, and 0.5 mg ÷ 0.01 = 50</i> <i>D is incorrect because 10 µg is 0.01 mg, and 0.5 mg ÷ 0.01 = 50</i>	(1)

Question number	Answer	Additional guidance	Mark
1(c)	An explanation that includes three of the following points: - not A and D because {they have no effect / the bacteria are resistant to them / there is no zone of inhibition} (1) - B as it {is most effective / has the largest zone of inhibition} (1) - use {high mass / 200 µg} antibiotic C as lower masses are not effective (1) - use antibiotic B at 10 µg as higher masses are no more effective (1)	ACCEPT use antibiotics B, C and E as {they are effective / the bacteria are not resistant to them / they have a zone of inhibition} PIECE TOGETHER ACCEPT do not use {low mass / 20µg} of C as it is not effective	(3)